

FarEye: GPS-Based Vehicle Tracking System for Schools

Gautam Kumar at Roboticwares has come up with a GPS plus SMS-based bus-tracking system that helps school authorities manage their school bus routes and keep an eye on their bus staff while saving the parents a lot of worry about bus timings



regarding bus timings for pick-up and drop.

When asked what really set off the spark for this idea in his head, Gautam answers, "We were looking into some of the problems faced by schools and parents with regards to running of the school buses. The issues affecting the former were generally regarding safety, finances and timing, while the latter were concerned specifically about safety and timings of pick-up and drop."

But aren't there similar systems already available in the market? Gautam clarifies, "There are normal GPS tracking systems available but

you are authorised to view) is shown on the top left corner of the screen. Simply choose the vehicle of your interest. You will be immediately shown all the details pertaining to it, viz, fuel level, speed, route, distance travelled and time spent stationary. The login page can also be opened on Android- and iOS-based mobile phones.

FarEye comes with its own GPS module along with an external antenna and u-blox GSM modem. The GPS is accurate up to ± 10 metres and the device comes with a replaceable warranty. The GPS module allows it to track bus movements very accurately and the on-board software maps the used routes to show you the location at any moment of time. The GSM modem is included to transmit to a server details like location, speed, stationary time and estimated time of arrival, where all this data is utilised to provide the functionalities described below.

Tracking of the movement. School authorities can track the movement of all the buses in real time anywhere in India. They can also keep track of time, stoppages and speed. Monitoring and tracking of buses improves day-to-day bus operations. It helps the school in identifying misused vehicles, unauthorised trips and entry into restricted areas.

Avoiding long wait and missing the school bus. Parents get automatic alerts via SMS a few minutes before the bus reaches their stop. It saves a long wait at the bus stops and even helps kids avoid missing the bus.

Graphs and extensive reports for analysis. Fuel reports and graphs are generated for a variety of parameters

ASHWIN GOPINATH

School buses are an integral part of a kid's life, but their unpredictable pick-up and drop timings are also one of the biggest causes of worry for their parents. Rarely do they come on time due to other students making them wait or delays on their routes.

Meet Gautam Kumar, 28, project director at Roboticwares, and his offering FarEye for Schools. FarEye, at its simplest, is a GPS plus SMS-based bus-tracking system that assists school authorities in managing the bus routes and keeping an eye on their bus staff while saving the parents a lot of worry

they don't address all the concerns of schools and parents. Furthermore, they provide tracking information over the computer, and it is not practical for parents to sit in front of a computer all day long tracking buses."

Working

The application opens up to a login page where you need to enter your credentials. You are



Login page

then taken to a screen that handles all the requests. A list of all vehicles powered by FarEye (and the ones